UNIT 3 · STANDARD 6.AT.3

Ratio Reasoning: Convert Measurements Between Systems

Lesson 3-7

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use ratio reasoning to convert measurements between the customary and metric systems.
Language Objective: I can explain a conversion between systems using the words conversion factor, approximately, customary, and metric.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Convert Measurements Between Systems | Conversion factor | Customary system | Metric system | Approximately | Ratio table

👆 *Tap any word to see what it means and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. Rewriting a customary measurement as a metric one gives an ________ answer, not an exact one.

2. The ratio that trades one unit for another is a ________ ________.

3. Inches, pounds and gallons belong to the ________ system.

4. Centimeters, grams and liters belong to the ________ system, built on tens.

5. Close to but not exactly equal is ________ — written with ≈.

6 A table whose every column holds the same comparison is a ______ ______.

Write About the Math

Is 100 kilometers per hour faster or slower than 100 miles per hour, and how do you know?

¿100 kilómetros por hora es más rápido o más lento que 100 millas por hora, y cómo lo sabes?

Say your answer to a partner first. Then write it, using at least two of these words:

- [ ] **Convert Measurements Between Systems** — *Convertir medidas entre sistemas*
- [ ] **Conversion factor** — *Factor de conversión*
- [ ] **Customary system** — *Sistema usual*
- [ ] **Metric system** — *Sistema métrico*
- [ ] **Approximately** — *Aproximadamente*

Model: *A mile is longer than a kilometer (1 km ≈ 0.6 mi), so 5 miles is the longer race even though the numbers match.* — Students reason about UNIT SIZE before touching arithmetic — same number, different units, different lengths.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The conversion factor is 1 km ≈ ____ mi. Scaling that up, 100 kilometers ≈ ____ miles, so the sign means about ____ miles per hour. That is ____ than 100 miles per hour, because in one hour you would travel ____ distance.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **My evidence is ____, so ____.**

  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.